

UNIT 1 - Class and Object

Problem Statement: Create Student class and display details.

Theory: A class is a blueprint and object is its instance.

Logic of Execution:

1. Create class
2. Add variables
3. Create object
4. Display

Screenshot of Code:

```
class Student { String name; int roll; float marks; }
```

Screenshot of Output:

Name: Rahul Roll:101 Marks:85.5

Conclusion: Demonstrates OOP basics.

UNIT 2 - Nested Class

Problem Statement: Create College and Department class.

Theory: Nested class is class inside another class.

Logic of Execution:

1. Create outer class
2. Create inner class
3. Display

Screenshot of Code:

```
class College { class Dept{} }
```

Screenshot of Output:

College and Department printed

Conclusion: Improves structure.

UNIT 3 - Exception Handling

Problem Statement: Handle multiple exceptions.

Theory: Exception handling prevents crash.

Logic of Execution:

1. Use try
2. Multiple catch

Screenshot of Code:

```
try{} catch{}
```

Screenshot of Output:

Error handled

Conclusion: Reliable program.

UNIT 4 - Vector

Problem Statement: Modify vector elements.

Theory: Vector is dynamic array.

Logic of Execution:

1. Add
2. Remove
3. Insert

Screenshot of Code:

Vector v = new Vector();

Screenshot of Output:

[HP, Dell, MSI...] Size=6

Conclusion: Dynamic structure.